Nonlinear Analysis, Theory, Methods & Applications, Vol. 21, No. 5, pp. 379–387, 1993.
Printed in Great Britain.

0362-546X/93 $6.00 + .00
© 1993 Pergamon Press Ltd

A FURTHER REMARK ON THE REGULARITY OF THE SOLUTIONS OF THE p-LAPLACIAN AND ITS APPLICATIONS TO THEIR FINITE ELEMENT APPROXIMATION

W. B. Liu and John W. Barrett
Department of Mathematics, Imperial College, London SW7 2BZ, U.K.

(*Received* 13 *February* 1992; *received for publication* 1 *March* 1993)

Key words and phrases: Error bounds, finite element approximation, p-Laplacian, regularity.

1. INTRODUCTION

Let Ω be a bounded open set in R^2 with a Lipschitz boundary $\partial\Omega$. Many mathematical models from physical processes have the following form (the p-Laplacian): $(1 < p < \infty)$

$$\begin{aligned} -\nabla \cdot (|\nabla u|^{p-2} \nabla u) &= f \qquad \text{in } \Omega \subset R^2 \\ u|_{\partial\Omega} &= g|_{\partial\Omega}, \end{aligned} \tag{1.1}$$

where the data f and g will be specified later. Such models arise in fluid mechanics (see [1, 2]), nonlinear diffusion (see [3]) and nonlinear elasticity (see [4]). We note that the case where $1 < p \leq 2$ is of the most interest is for elastic–plastic models. The well-posedness of (1.1) has been proved by the monotonicity method (see [5, 6]); there being a unique solution in $W^{1,p}(\Omega)$. Unfortunately it is impossible to give its explicit solution except in some very special cases (see [1, 4]). Therefore, its numerical approximation is of great importance. Some error bounds for finite element approximation of this equation have been established; see, for example, [5–9]. We summarize these results below. Let Ω be a polygonal domain and S^h be the space of continuous piecewise linear functions based on a regular triangulation of Ω. The finite element approximation of (1.1) from S^h is then: find $u^h \in V_g^h \equiv \{v^h \in S^h : v^h|_{\partial\Omega} = g^h\}$, where g^h is an appropriate approximation of g on $\partial\Omega$, such that

$$\int_\Omega |\nabla u^h|^{p-2}(\nabla u^h, \nabla v^h) = \int_\Omega f v^h \qquad \forall\, v^h \in V_0^h. \tag{1.2}$$

Throughout this paper $(\cdot, \cdot)$ denotes the R^2 inner product.

The well-posedness of (1.2) is easily established and the existing error bounds can be stated as follows: for any $v^h \in V_g^h$

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leq \begin{cases} C[\|u - v^h\|_{W^{1,p}(\Omega)}]^{p/2} & p \leq 2 \\ C[\|u - v^h\|_{W^{1,p}(\Omega)}]^{2/p} & p \geq 2, \end{cases} \tag{1.3}$$

where C depends on $\|v^h\|_{W^{1,p}(\Omega)}$ for $p > 2$ (see [8]). Therefore, if $u \in W^{2,p}(\Omega)$, it follows that

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leq \begin{cases} Ch^{p/2} & p \leq 2 \\ Ch^{2/p} & p \geq 2. \end{cases} \tag{1.4}$$

Recently improved error bounds have been established in [7, 9], which can be stated as follows: for any $v^h \in V_g^h$

$$\|u - u^h\|^2_{W^{1,p}(\Omega)} \leq C\|u - v^h\|^2_{(p,\delta)} \qquad \text{if } p \leq 2 \text{ and } -2 < \delta \leq 0 \tag{1.5}$$

and

$$\|u - u^h\|^2_{(p,\delta)} \leq C\|u - v^h\|^2_{W^{1,p}(\Omega)} \qquad \text{if } p \geq 2 \text{ and } \delta \geq 0, \tag{1.6}$$

where $\|v\|^2_{(p,\delta)} \equiv \int_\Omega (|\nabla u| + |\nabla v|)^{p-2-\delta}|\nabla v|^{2+\delta}$ with $\delta \geq -2$. These seem the best existing results and specifically one can prove from them that

$$\|u - u^h\|_{W^{1,p}(\Omega)} \leq Ch \tag{1.7}$$

if $1 < p < 2$ and $u \in C^{2,(2-p)/p}(\bar{\Omega}) \cap W^{3,1}(\Omega)$ or $u \in W^{3,p}(\Omega)$, and under different assumptions optimal error bounds for $p > 2$.

Nevertheless, all the existing error bounds depend on the regularity of u. For example, it is well known that u is rarely in $W^{2,p}(\Omega)$ even if f and g are very smooth when $p > 2$ so that (1.3) yields convergence rates much poorer than $2/p$. However, we note from (1.6) that it is a simple matter to deduce (1.4) for $p \geq 2$ by requiring $u \in H^2(\Omega) \cap W^{1,\infty}(\Omega)$ only, see [9]. Similarly, if one does not know the regularity of u, one can deduce little from (1.5) and (1.6). It is, therefore, necessary to obtain some regularity results if one wishes to obtain any explicit error bounds from these abstract bounds.

Recently there has been much work on the regularity of solutions of (1.1) and one can refer to [10–17] for such work. However, there appears to be no existing regularity results that can be applied here, except those in [13], as all the other regularity results have either a first order or local nature.

In [13] global H^2 regularity is established for the solutions of the p-Laplacian equation, $1 < p \leq 2$, and related equations on domains which are either convex or have C^2 boundaries. Then, these results are used to derive some explicit error bounds for the finite element approximation of the p-Laplacian from (1.3), which were not previously available because of the lack of regularity results. Clearly these regularity results are not sufficient to establish optimal error bounds from (1.5). On the other hand it should be noted that the regularity requirements in [7, 9] to obtain (1.7) from (1.5) appear to be suboptimal; for example if $p = 2$ one sees immediately that the regularity requirements are, at least in this case, too high. Therefore, it is important to establish whether they can be further reduced.

The purpose of this short note is twofold. The first is to show that the regularity requirement can be further reduced to obtain (1.7) from (1.5). The second purpose is to prove that these regularity conditions can indeed be fulfilled in some physically relevant cases. Therefore, in these cases the piecewise linear finite element approximation of the p-Laplacian does converge at the optimal rate in the $W^{1,p}$ norm.

Throughout this paper, we adopt the standard notation $W^{s,q}(\Omega)$ ($s \geq 0$, $q \in [1, \infty]$) for Sobolev spaces on Ω with norm $\|\cdot\|_{W^{s,q}(\Omega)}$. We denote $W^{m,2}(\Omega)$ by $H^m(\Omega)$ for nonnegative integers m. For the definitions of the spaces $C^{k,\alpha}(\Omega)$ (k a nonnegative integer and $0 \leq \alpha \leq 1$) and domains with $C^{k,\alpha}$ boundaries, see [18] for example. The spaces $W^{s,q}_{\text{loc}}(\Omega)$ consist of all functions which are in $W^{s,q}(G)$ for all open sets $G \subset\subset \Omega$ (that is $\bar{G} \subset \Omega$). In addition C denotes a general positive constant independent of h.

2. FINITE ELEMENT APPROXIMATION AND ABSTRACT ERROR BOUNDS

In this section we consider the piecewise linear finite element approximation of (1.1) and try to weaken the regularity conditions required in [9] to obtain (1.7) from (1.5). We achieve this by exploring some regularity properties of the solutions of (1.1). The weak formulation of (1.1) is

(WP) find $u \in V_g^p = \{v \in W^{1,p}(\Omega)\colon v|_{\partial\Omega} = g\}$ such that

$$\int_\Omega |\nabla u|^{p-2}(\nabla u, \nabla v) = \int_\Omega fv \qquad \forall\, v \in V_0^p,$$

where $f \in (W^{1,p}(\Omega))^*$ (the dual space of $W^{1,p}(\Omega)$). Associated with (WP) is the following minimization problem:

(MP) find $u \in V_g^p$ such that

$$J_\Omega(u) = \operatorname*{Min}_{v \in V_g^p} J_\Omega(v),$$

where

$$J_\Omega(v) \equiv \int_\Omega \int_0^{|\nabla v|} t^{p-1}\,\mathrm{d}t - \int_\Omega fv. \tag{2.1}$$

It is not difficult to see that J_Ω is well defined and continuous on V_g^p. Moreover, J_Ω is Gateaux differentiable with $J'_\Omega(u)(v) \equiv \int_\Omega |\nabla u|^{p-2}(\nabla u, \nabla v) - \int_\Omega fv$. In the following, we will always assume that $f \in L^2(\Omega)$. A solution of (WP) will be called a weak solution of (1.1). The well-posedness of (WP) has been well established. Furthermore, J_Ω is strictly convex and coercive on V_g^p so that J_Ω has a unique minimizer on V_g^p, which satisfies its Euler equation (WP). In other words the problems (WP) and (MP) are equivalent. All these conclusions can be found in [5, 6, 8]. As explained in [13], we will mainly deal with the case $1 < p \leq 2$ and in the rest of the paper we will suppose this unless otherwise stated.

Consider the finite element approximation of (WP). Let Ω^h be a polygonal approximation to Ω with boundary $\partial\Omega^h$. Let T^h be a partitioning of Ω^h into disjoint open regular triangles τ, each of maximum diameter bounded above h, so that $\bar{\Omega}^h \equiv \bigcup_{\tau \in T^h} \bar{\tau}$. We assume that $P_i \in \partial\Omega^h \Rightarrow P_i \in \partial\Omega$ and $\operatorname{dist}(\partial\Omega^h, \partial\Omega) \leq Ch^2$, where $\{P_i\}$ $(i = 1 \to I)$ is the vertex set associated with the triangulation T^h.

Associated with T^h is a finite dimensional subspace S^h of $C^0(\bar{\Omega}^h)$ such that $\chi_{|_\tau}$ is linear for all $\chi \in S^h$ and $\tau \in T^h$. For ease of exposition we will assume that $\Omega^h \subseteq \Omega$. Let $\pi_h\colon C(\bar{\Omega}^h) \to S^h$, denote the interpolation operator such that for any $v \in C(\bar{\Omega}^h)$, $(\pi_h v)(P_i) = v(P_i)$ $i = 1 \to I$. Let $V_g^h \equiv \{\chi \in S^h\colon \chi(P_i) = g(P_i)$ for any $P_i \in \partial\Omega\}$. It follows that $V_g^h \subset W^{1,p}(\Omega^h)$. A possible finite element approximation of (1.1) is then

$(\mathrm{MP})^h$ find $u^h \in V_g^h$ such that

$$J_{\Omega^h}(u^h) = \operatorname*{Min}_{v^h \in V_g^h} J_{\Omega^h}(v^h),$$

where $J_{\Omega^h}(\cdot)$ is given by (2.1) with Ω replaced by Ω^h. Equivalently one can define u^h as the unique solution of

$(\mathrm{WP})^h$ find $u^h \in V_g^h$ such that

$$\int_{\Omega^h} |\nabla u^h|^{p-2}(\nabla u^h, \nabla v^h) = \int_{\Omega^h} fv^h \qquad \forall\, v^h \in V_0^h.$$

The well-posedness and equivalence of the above is also well established, see [5, 6, 8]. In [7, 9], $(1.5)^h$ and $(1.6)^h$ were established and it was shown that $(1.3)^h$ could easily be obtained from them (where $(1.5)^h$ denotes (1.5) with Ω replaced by Ω^h, etc.). Furthermore, it was proved that $(1.7)^h$ is true if $u \in C^{2,(2-p)/p}(\bar{\Omega}) \cap W^{3,1}(\Omega)$ or $u \in W^{3,p}(\Omega)$. In this section we will show that the same conclusion can be achieved if $u \in W^{1+2/p,p}(\Omega)$ provided $f \in C^\gamma(\bar{\Omega}\backslash\Gamma)$ $(\gamma > 0)$, where Γ consists of finite piecewise-smooth curves in Ω, $g = 0$, $\partial\Omega \in C^2$ and T^h is quasi-uniform.

THEOREM 2.1. Let $p \in (1, 2)$ and let $f \in C^\gamma(\bar{\Omega}\backslash\Gamma)$ with $\gamma > 0$ and Γ consisting of finite piecewise-smooth curves. Let T^h be quasi-uniform. If $u \in W^{1+2/p,p}(\Omega)$ and

$$\int_\Omega |\nabla u|^{p-2}|H[u]|^2 < \infty,$$

where $H[u]$ is the Hessian matrix of u, then

$$\|u - u^h\|_{W^{1,p}(\Omega^h)} \leq Ch. \tag{2.2}$$

Proof. Firstly, we note the imbedding $u \in W^{1+2/p,p}(\Omega) \Rightarrow u \in H^2(\Omega)$. From [17] we have that u solving (WP) $\Rightarrow u \in C^{1,\alpha}(\Omega)$ for some $\alpha > 0$. Let $D \equiv \{x \in \Omega: |\nabla u(x)| > 0\}$, an open set in Ω. We have that

$$-\Delta u - (p - 2)\sum_{i,j}^{2} (u_{x_i}u_{x_j}/|\nabla u|^2)u_{x_ix_j} = |\nabla u|^{2-p}f \qquad \text{in } D\backslash\Gamma, \tag{2.3}$$

a uniformly elliptic equation with coefficients and right-hand side Holder continuous over $D\backslash\Gamma$. From the classical elliptic partial differential equation theory, e.g. see theorem 9.19 in [18], it follows that $u \in C^2(D\backslash\Gamma)$ so that $H[u]$ is continuous on $D\backslash\Gamma$. On the other hand, $|H[u]| = 0$ almost everywhere on $\Omega\backslash D$. Hence, $H[u]$ is continuous almost everywhere on Ω.

Let $R^h \subset\subset \Omega$ be such that $\text{mes}(\Omega\backslash R^h) \leq Ch^4$. Therefore, for every triangle $\tau_k \in T^h$ there exists a $z_k \in r_k \equiv \tau_k \cap R^h$, a continuous point of $H[u]$, such that for almost all $x \in r_k$

$$|\nabla(u - \pi_h u)(x)| \leq Ch|H[u](z_k)| + Ch^2 \leq Ch|H[u](x)| + Ch|H[u](x) - H[u](z_k)| + Ch^2,$$

since $u \in C^1(\bar{r}_k)$. A simple calculation, as in [7, 9], yields that

$$\begin{aligned}
\|u - \pi_h u\|^2_{(p,0)} &\equiv \int_{\Omega^h} (|\nabla u| + |\nabla(u - \pi_h u)|)^{p-2}|\nabla(u - \pi_h u)|^2 \\
&\leq \int_{R^h} (|\nabla u| + |\nabla(u - \pi_h u)|)^{p-2}|\nabla(u - \pi_h u)|^2 + \int_{\Omega^h\backslash R^h} |\nabla(u - \pi_h u)|^p \\
&\leq Ch^2 \int_{R^h} |\nabla u|^{p-2}|H[u]|^2 + Ch^p \sum_k \int_{r_k} |H[u](x) - H[u](z_k)|^p \, dx \\
&\quad + C[\text{mes}(\Omega^h\backslash R^h)]^{1/2}\|u\|^p_{H^2(\Omega)} + Ch^{2p}.
\end{aligned} \tag{2.4}$$

Let $s = (2 - p)/p$, so that $1 + 2/p = 2 + s$ with $s \in (0, 1)$. Let

$$M^h \equiv \max_k \left\{ [\mathrm{mes}(\tau_k)]^{-1} \int_{\tau_k} [|H[u](x) - H[u](z_k)|^p / h^{2+ps}] \, \mathrm{d}x \right\}$$

and

$$[t]_{M^h} \equiv \begin{cases} M^h & \text{if } t \geq M^h \\ t & \text{if } t \leq M^h. \end{cases}$$

Noting that T^h is quasi-uniform we have that $\mathrm{mes}(\tau_k) \geq Ch^2$ and so

$$\begin{aligned} h^p \sum_k \int_{\tau_k} & |H[u](x) - H[u](z_k)|^p \, \mathrm{d}x \\ & \leq Ch^{p(1+s)} \sum_k \mathrm{mes}(\tau_k) \int_{\tau_k} [|H[u](x) - H[u](z_k)|^p / h^{2+ps}] \, \mathrm{d}x \\ & \leq Ch^2 \sum_k [\mathrm{mes}(\tau_k)]^2 \operatorname*{ess\,sup}_{(x,z) \in \tau_k \times \tau_k} [|H[u](x) - H[u](z)|^p / |x - z|^{2+ps}]_{M^h} \\ & \leq Ch^2 \sum_k \sum_l \mathrm{mes}(\tau_k) \, \mathrm{mes}(\tau_l) \operatorname*{ess\,sup}_{(x,z) \in \tau_k \times \tau_l} [|H[u](x) - H[u](z)|^p / |x - z|^{2+ps}]_{M^h} \\ & \leq Ch^2 \|u\|^p_{W^{1+2/p,p}(\Omega)}. \end{aligned} \tag{2.5}$$

Combining $(1.5)^h$, (2.4) and (2.5) yields that

$$\|u - u^h\|^2_{W^{1,p}(\Omega^h)} \leq Ch^2 \int_\Omega |\nabla u|^{p-2} |H[u]|^2 + Ch^2 \|u\|^p_{W^{1+2/p,p}(\Omega)} + Ch^2$$

and, hence, the desired result (2.2) under the given assumptions. ■

Remark 2.1. We note that the regularity requirements on u in theorem 2.1 above collapse to $H^2(\Omega)$ for $p = 2$, the minimal regularity requirement for (2.2) to hold in this case. If $u \in W^{3,1}(\Omega)$, it follows from lemma 5.1 in [9] that the integral $\int_\Omega |\nabla u|^{p-2} |H[u]|^2$ is finite so that the conclusion of theorem 2.1 remains true if $u \in W^{3,1}(\Omega) \cap W^{1+2/p,p}(\Omega)$. But this regularity does not collapse to $H^2(\Omega)$ when $p = 2$. In many cases, however, one can show that this integral is finite without imposing extra regularity conditions on u. To illustrate this we consider the case $g = 0$ and $\partial\Omega \in C^2$.

THEOREM 2.2. Let $p \in (1, 2)$, $f \in L^\infty(\Omega)$ and $u \in W^{2,p}(\Omega)$; then for any $S \subset\subset \Omega$

$$\int_S |\nabla u|^{p-2} |H[u]|^2 < \infty. \tag{2.6}$$

If in addition $\partial\Omega \in C^2$ and $g = 0$, then

$$\int_\Omega |\nabla u|^{p-2} |H[u]|^2 < \infty. \tag{2.7}$$

If in addition to the above assumptions $u \in W^{1+2/p,p}(\Omega)$, $f \in C^\gamma(\bar{\Omega} \backslash \Gamma)$ with $\gamma > 0$ and Γ consisting of finite piecewise-smooth curves, and T^h is quasi-uniform; then (2.2) holds.

Proof. We first show that $\int_{B_R(x)} |\nabla u|^{p-2} |H[u]|^2 < \infty$ for any open ball $B_R(x)$ such that $B_{2R}(x) \subset\subset \Omega$. We note that the p-Laplacian satisfies the structural conditions (1.3)–(1.7) with $\kappa = 0$ in [17]. From (2.2.4) in [17], with G being the identity map and $\psi \in C_0^\infty(\Omega)$ such that $\psi \geq 0$ and $\psi = 1$ on $B_R(x)$, and the structural inequality (1.4) (note there is a misprint in [17]: $p - 1$ should be replaced by $p - 2$) one easily deduces that the above integral is finite as $u \in W^{2,p}(\Omega)$ and $f \in L^\infty(\Omega)$. Hence, (2.6) holds.

As $g = 0$ and $\partial\Omega \in C^2$, one can use the reflection method (see [16, 19]) to show that for any $x \in \partial\Omega$ u can be extended to satisfy an equation with similar structural conditions in a ball $B_R(x)$ with extended data $f \in L^\infty(B_R(x))$. From [17] it follows that $u \in C^{1,\alpha}(B_R(x))$ for some $\alpha > 0$ and, hence, from the reflection construction that $u \in W^{2,p}(B_R(x))$. Consequently the desired integral is finite over $B_R(x)$ for any $x \in \partial\Omega$ and, therefore, (2.7) follows.

Finally the result (2.2) follows immediately from theorem 2.1. ■

Remark 2.2. We believe that (2.7) remains true for $g \in W^{1+2/p,p}(\Omega)$ (see also the proof of theorem 3.3 below). If this was true (we have no proof at present) the only essential regularity requirement would be that $u \in W^{1+2/p,p}(\Omega)$, which collapses to $H^2(\Omega)$ when $p = 2$.

Remark 2.3. Theorems 2.1 and 2.2 remain true for more general equations of the type

$$-\nabla \cdot (k(x, |\nabla u(x)|)\, \nabla u(x)) = f(x) \qquad \forall\, x \in \Omega \subset R^2, \tag{2.8}$$

where $(x, t) \in \bar{\Omega} \times (0, \infty) \to k(x, t)$ is smooth with respect to x and satisfies the assumptions (A) in [9] with respect to t, these are very similar to the assumptions in [8] for $p \in (1, 2)$. This class of equations includes the p-Laplacian and many other physically relevant problems.

3. REGULARITY OF THE SOLUTIONS OF THE p-LAPLACIAN

In the previous two sections one sees that it is essential to know some regularity of u, the solution of (1.1), to obtain any explicit error bounds. From [13] we have the following result.

THEOREM 3.1. Let $p \in (1, 2)$, $\partial\Omega \in C^2$, $f \in L^q(\Omega)$ $(q > 2)$ and $g \in H^2(\Omega)$ then $u \in H^2(\Omega)$. ■

From this result and $(1.3)^h$ one can readily obtain $(1.4)^h$, see [13]. However, it is not possible to obtain (2.2) from this regularity result. Although some higher local regularity is known (see theorem 3.2 below), for (2.2) to hold we require higher global regularity results. Below we show that the regularity required in theorem 2.1 can indeed be achieved in some physically relevant cases.

Consider the following mechanics problem: suppose a plane Ω, which is made of a material obeying the power law with $p \in (1, 2)$, is undergoing longitudinal shear (the mode III problem, see [4]). In this case the only nonvanishing component of displacement $u(x)$ is normal to Ω and satisfies the p-Laplacian for $x \in \Omega$. Physically one expects the tangential strain of the plane along $\partial\Omega$ will vary strictly monotonically except at those points where the displacement $u(x)$ achieves its extreme values. Motivated by this observation we will suppose that the given boundary data g in (1.1) has a nonvanishing tangential derivative, g_τ, on $\partial\Omega$ except at those points where g achieves its global extrema on $\partial\Omega$. Under this assumption we will show that u, the solution of (1.1), satisfies the regularity requirements of theorem 2.1. First we quote a known result, see [11].

THEOREM 3.2. Let $u \in W^{1,p}(\Omega)$, $1 < p < \infty$, be the solution of (WP) with $f = 0$, then

$$u \in C^{k,\beta}(\Omega) \cap W^{k+2,q}_{\text{loc}}(\Omega), \tag{3.1}$$

where the integer $k \geq 1$ and exponents $\beta \in (0, 1]$ and $q \in [1, 2]$ are determined uniquely from

$$6(k + \beta) = 7 + 1/(p - 1) + [1 + 14/(p - 1) + 1/(p - 1)^2]^{1/2} \quad \text{and} \quad 1 \leq q < 2/(2 - \beta). \tag{3.2}$$

Note that the above is not necessarily true if $f \neq 0$ (e.g. consider the case f being a constant). We now prove a global regularity result.

THEOREM 3.3. Let $p \in (1, 2)$, $f = 0$ and $\partial\Omega \in C^2$. Let $g \in C^{1,\gamma}(\bar{\Omega}) \cap H^2(\Omega)$ $(\gamma > 0)$ be such that its tangential derivative along $\partial\Omega$, g_τ, satisfies $|g_\tau(x_0)| > 0$ if $x_0 \in \partial\Omega$ and $g(x_0)$ is not a global extremum of g on $\partial\Omega$. Then (2.7) holds.

Furthermore if $\partial\Omega \in C^{2,1}$, then for $\gamma > 0$ it follows that:

(i) $g \in C^{2,\gamma}(\bar{\Omega}) \Rightarrow u \in C^{2,\mu}(\bar{\Omega})$;

(ii) $g \in C^{2,\gamma}(\bar{\Omega}) \cap W^{3,r}(\Omega)$, $r \in (1, p] \Rightarrow u \in C^{2,\mu}(\bar{\Omega}) \cap W^{3,r}(\Omega)$;

where $\mu = \gamma$ for $p \in (1, 1.3)$, $\mu = \min(\gamma, \beta)$ for $p \in [1.3, 2)$ and

$$\beta = \{-5 + 1/(p - 1) + [1 + 14/(p - 1) + 1/(p - 1)^2]^{1/2}\}/6 \geq (2 - p)/p. \tag{3.3}$$

Proof. From [12] we have that $g \in C^{1,\gamma}(\bar{\Omega}) \Rightarrow u \in C^{1,\alpha}(\bar{\Omega})$ for some $\alpha > 0$, so that $|\nabla u|$ is continuous on $\bar{\Omega}$. We first show that $|\nabla u|_{\partial\Omega}| \geq \eta > 0$. If $x \in \partial\Omega$ and $g_\tau(x) \neq 0$, then $|\nabla u(x)| > 0$ as $(u - g)|_{\partial\Omega} = 0$. Let $g_\tau(x_0) = 0$ and (without loss of generality as $f = 0$) let $g(x) \geq g(x_0)$ for all $x \in \partial\Omega$. Let $u_0 = g(x_0)$. Then u_0 is such that $-\nabla \cdot (|\nabla u_0|^{p-2} \nabla u_0) = 0$ in Ω with $(u - u_0)|_{\partial\Omega} \geq 0$. If u is constant over Ω, then the desired regularity result holds. Therefore, we can assume that u is not constant. From the strong maximum principle in [16, 20] it follows that $u > u_0$ in Ω. From the Hopf maximum principle, see [16] and lemma 3.4 in [18], we have that $|\partial u/\partial n(x_0)| > 0$, where n is the outward unit normal to $\partial\Omega$. Consequently $|\nabla u|_{\partial\Omega}| > 0$ and from the continuity of $|\nabla u|$, there exists a domain $S \subset\subset \Omega$ such that $\partial S \in C^{2,1}$ and $|\nabla u|_{\bar{\Omega}\setminus S}| \geq \eta > 0$. It follows from theorem 3.1 that $g \in H^2(\Omega) \Rightarrow u \in H^2(\Omega)$ and, hence, $\int_{\Omega\setminus S} |\nabla u|^{p-2} |H[u]|^2$ is finite. On the other hand from (2.6) it follows that $\int_S |\nabla u|^{p-2} |H[u]|^2$ is finite. Therefore, the desired result (2.7) holds if $g \in C^{1,\gamma}(\bar{\Omega}) \cap H^2(\Omega)$.

Let S_1 be such that $S \subset\subset S_1 \subset\subset \Omega$, then from theorem 3.2 we have that

$$p \in (1, 1.3) \Rightarrow k \geq 3 \Rightarrow u \in C^3(\bar{S}_1),$$

$$p \in [1.3, 2) \Rightarrow k = 2 \Rightarrow u \in C^{2,\beta}(\bar{S}_1) \cap W^{3,p}(S_1) \text{ with } \beta \text{ given by (3.3).} \tag{3.4}$$

Rewriting (WP) as in (2.3), we have for $Q \equiv \Omega\setminus S$ and $g \in C^{2,\gamma}(\bar{\Omega})$ that

$$\Delta u + (p - 2) \sum_{i,j}^{2} (u_{x_i} u_{x_j} / |\nabla u|^2) u_{x_i x_j} = 0 \text{ in } Q \quad \text{and} \quad u|_{\partial Q} \in C^{2,\mu}(\partial Q).$$

As $(u_{x_i} u_{x_j} / |\nabla u|^2) \in C^{\alpha}(\bar{Q})$ it follows from theorem 9.19 in [18], for example, that $u \in C^{2,\sigma}(\bar{Q})$, $\sigma = \min(\alpha, \mu)$, yielding $(u_{x_i} u_{x_j} / |\nabla u|^2) \in C^{1,\sigma}(\bar{Q})$ and, hence, $u \in C^{2,\mu}(\bar{Q})$. Therefore, it follows that $u \in C^{2,\mu}(\bar{\Omega})$. If $g \in C^{2,\gamma}(\bar{\Omega}) \cap W^{3,r}(\Omega)$, $r \in (1, p]$, it follows in a similar way that $u \in C^{2,\mu}(\bar{\Omega}) \cap W^{3,r}(\Omega)$. ■

Therefore, under the assumptions of theorem 3.3 with $g \in C^{2,(2-p)/p}(\bar{\Omega}) \cap W^{3,r}(\Omega)$, $r \in (1, p]$, one has $u \in C^{2,(2-p)/p}(\bar{\Omega}) \cap W^{3,r}(\Omega)$. Then one can apply the result in [7] to establish (2.2). Similarly if $g \in C^{2,\gamma}(\bar{\Omega}) \cap W^{3,p}(\Omega)$, $\gamma > 0$, one has $u \in W^{3,p}(\Omega)$ and one can apply the result in [9] to establish (2.2). Below we establish (2.2) under the assumption $g \in C^{2,(2-p)/p}(\bar{\Omega})$ by applying theorems 2.1 and 3.3 as follows.

THEOREM 3.4. Let $p \in (1, 2)$, $f = 0$ and $\partial\Omega \in C^{2,1}$. Let $g \in C^{2,(2-p)/p}(\bar{\Omega})$ be such that $|g_\tau(x_0)| > 0$ if $x_0 \in \partial\Omega$ and $g(x_0)$ is not a global extremum of g on $\partial\Omega$. Let T^h be quasi-uniform, then (2.2) holds.

Proof. From theorem 3.3 it follows that (2.7) holds and $u \in C^{2,(2-p)/p}(\bar{\Omega})$. The latter implies that $u \in W^{1+2/p,p}(\Omega)$. Therefore, the desired result follows from theorem 2.1. ■

Remark 3.1. As stated previously in remark 2.2 we believe that $g \in W^{1+2/p,p}(\Omega)$ should be sufficient to prove (2.7). In addition we believe it should also be sufficient to prove that $u \in W^{1+2/p,p}(\Omega)$ and, hence, theorem 3.4.

Remark 3.2. Theorems 3.2–3.4 also hold for the more general equation (2.8) discussed in remark 2.3 above. The key point is to show that the high order local regularity result, theorem 3.2, still holds for such equations and this is proved in [14].

Acknowledgements—The authors wish to express their sincere thanks to their colleague Dr C. R. Champion for many useful discussions on the mechanics model. The financial support of the first author by the SERC under grant GR/F81255 is gratefully acknowledged.

REFERENCES

1. ATKINSON C. & JONES C. W., Similarity solutions in some nonlinear diffusion problems and in boundary-layer flow of a pseudo plastic fluid, *Q. J. Mech. Appl. Math.* **27**, 193–211 (1974).
2. BARANGER J. & NAJIB K., Numerical analysis for quasi-Newtonian flow obeying the power or the Carreau law, *Num. Math.* **58**, 35–49 (1990).
3. PHILIP J. R., *N*-diffusion, *Aust. J. Phys.* **14**, 1–13 (1961).
4. ATKINSON C. & CHAMPION C. R., Some boundary value problems for the equation $\nabla \cdot (|\nabla\phi|^N \nabla\phi) = 0$, *Q. J. Mech. Appl. Math.* **37**, 401–419 (1984).
5. CIARLET P. G., *The Finite Element Method for Elliptic Problems*. North Holland, Amsterdam (1978).
6. GLOWINSKI R. & MARROCCO A., Sur l'approximation par elements finis d'order un, et la resolution, par penalisation-dualite, d'une classe de problemes de Dirichlet non lineaires, *R.A.I.R.O. Analyse Numerique* **2**, 41–64 (1975).
7. BARRETT J. W. & LIU W. B., Finite element approximation of the *p*-Laplacian, *Math. Comp.* (1993) (to appear).
8. CHOW S. S., Finite element error estimates for nonlinear elliptic equations of monotone type, *Num. Math.* **54**, 373–393 (1989).
9. LIU W. B. & BARRETT J. W., Finite element approximation of some degenerate monotone quasilinear elliptic systems (submitted for publication).
10. DIBENEDETTO E., $C^{1+\alpha}$ local regularity of weak solutions of degenerate elliptic equations, *Nonlinear Analysis* **7**, 827–850 (1983).
11. IWANIEC T. & MANFREDI J. J., Regularity of *p*-harmonic functions on the plane, *Rev. Math. Iberoamericana* **5**, 1–19 (1989).
12. LIEBERMAN G. M., Boundary regularity for solutions of degenerate elliptic equations, *Nonlinear Analysis* **12**, 1203–1219 (1988).
13. LIU W. B. & BARRETT J. W., A remark on the regularity of the solutions of the *p*-Laplacian and its applications to their finite element approximation, *J. math. Analysis Applic.* (1993) (to appear).
14. LIU W. B. & BARRETT J. W., Higher order regularity for the solutions of some degenerate quasilinear elliptic equations in the plane, *SIAM J. math. Analysis* (1993) (to appear).

15. THELIN F. DE, Local regularity properties for the solutions of a nonlinear partial differential equation, *Nonlinear Analysis* **6**, 839–844 (1982).
16. TOLKSDORF P., On the Dirichlet problem for quasilinear equations in domains with conical boundary points, *Communs partial diff. Eqns* **8**, 773–817 (1983).
17. TOLKSDORF P., Regularity for a more general class of quasilinear elliptic equations, *J. diff. Eqns* **51**, 126–150 (1984).
18. GILBARG D. & TRUDINGER N., *Elliptic Partial Differential Equations of Second Order*, 2nd edition. Springer, Berlin (1984).
19. GUEDDA M. & VERON L., Quasilinear elliptic equations involving critical Sobolev exponents, *Nonlinear Analysis* **13**, 879–902 (1989).
20. MANFREDI J. J., p-Harmonic functions in the plane, *Proc. Am. math. Soc.* **103**, 473–479 (1988).